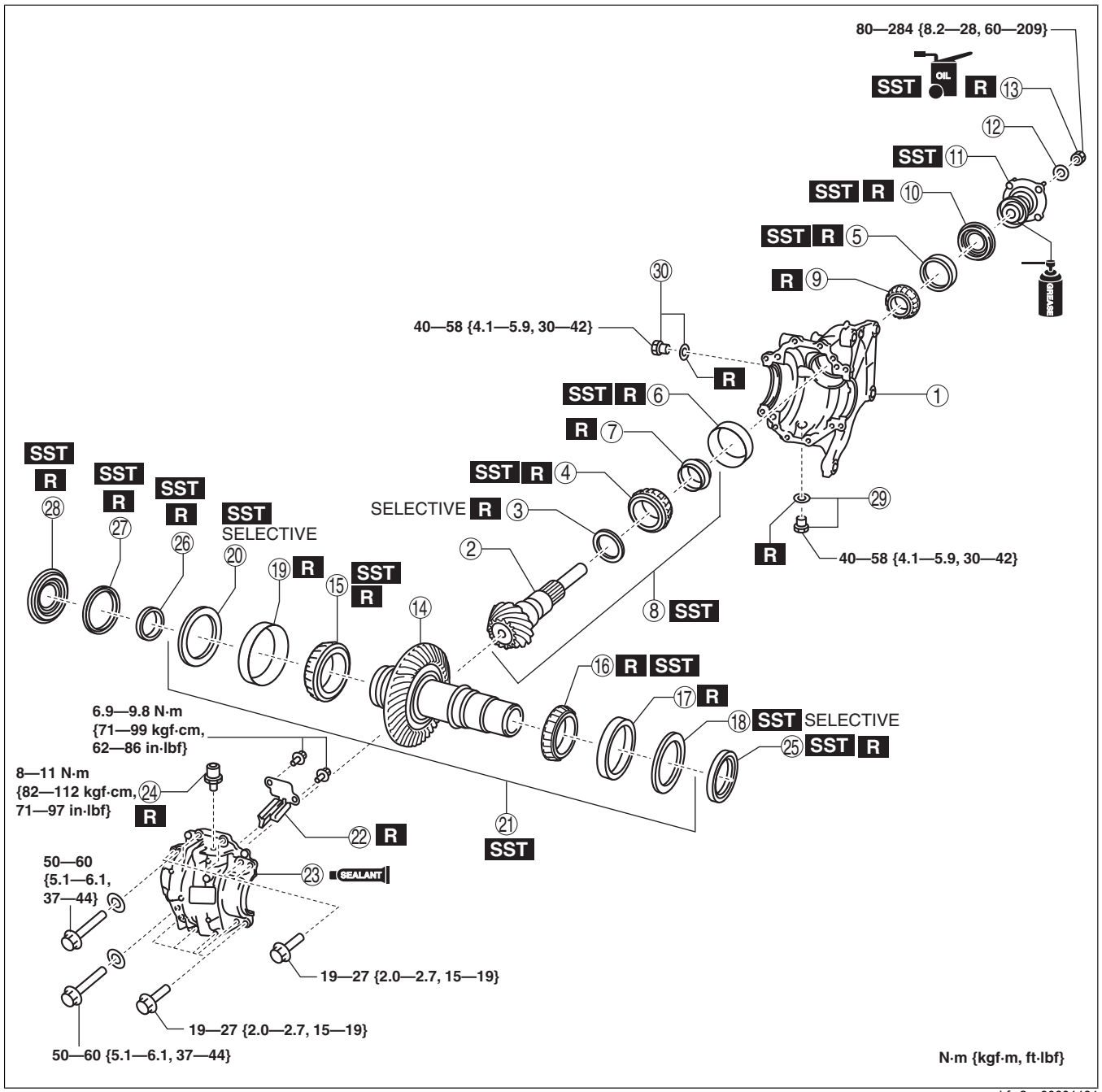
TRANSFER ASSEMBLY [EW6AX-EL]

id0316h2000300

Caution

- **To prevent the silicone sealant from hardening, adhere the drive gear case to the front carrier within 3 min after applying silicone sealant.**
- **Wait 30 min or more after the silicone sealant application area has bonded and fill with transfer oil.**



bfw2za00001131

1	Front carrier
2	Drive pinion gear
3	Spacer
4	Bearing inner race (front)
5	Bearing outer race (rear)
6	Bearing outer race (front)
7	Collapsible spacer
8	Drive pinion gear component
9	Bearing inner race (rear)
10	Oil seal

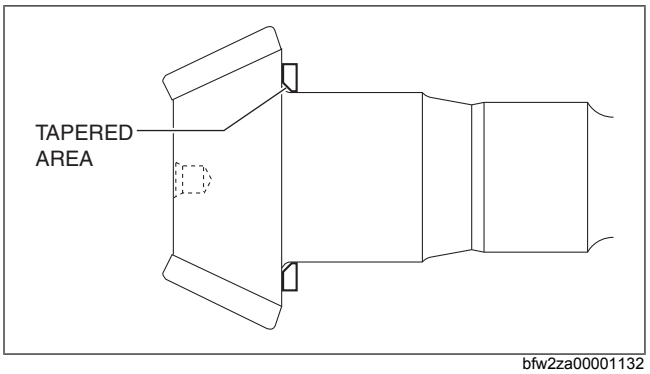
11	Companion flange component
12	Washer
13	Locknut
14	Ring gear shaft
15	Bearing inner race (RH)
16	Bearing inner race (LH)
17	Bearing outer race (LH)
18	Adjustment shim (LH)
19	Bearing outer race (RH)
20	Adjustment shim (RH)
21	Ring gear shaft component
22	Baffle plate
23	Drive gear case
24	Breather
25	Transfer oil seal (LH)
26	Transfer oil seal (RH) No.3
27	Transfer oil seal (RH) No.2
28	Transfer oil seal (RH) No.1
29	Drain plug, washer
30	Oil level plug, washer

1. Clean the contact surfaces of the front carrier and drive gear case.

Note
- Perform drive pinion preload adjustment with the oil seal not assembled.

2. Assemble the drive pinion gear component and companion flange to the front carrier using the following procedure.
   (1) When disassembling, measure the thickness of the removed spacer and assemble a new spacer with the same dimensions to the drive pinion gear.

Note
- Verify the assembly direction of the spacer and assemble it.



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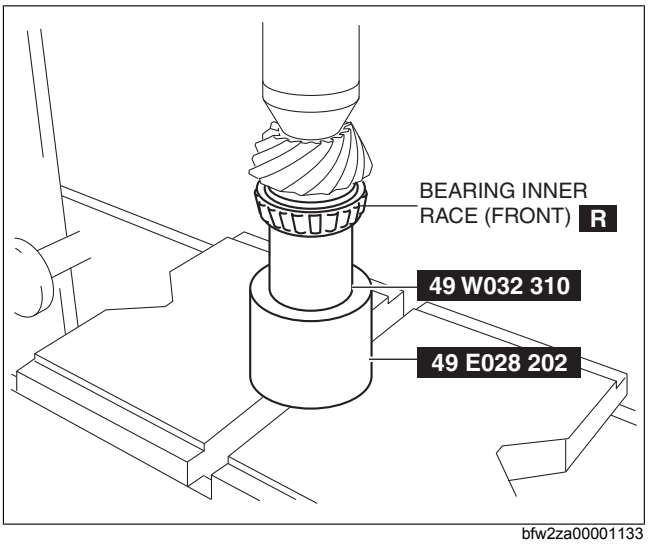
Spacer table

Identification number	Thickness (mm {in})	Identification number	Thickness (mm {in})
08	3.080 {0.1213}	29	3.290 {0.1295}
09	3.095 {0.1219}	30	3.305 {0.1301}
11	3.110 {0.1224}	32	3.320 {0.1307}
12	3.125 {0.1230}	33	3.335 {0.1313}
14	3.140 {0.1236}	35	3.350 {0.1319}
15	3.155 {0.1242}	36	3.365 {0.1325}
17	3.170 {0.1248}	38	3.380 {0.1331}
18	3.185 {0.1254}	39	3.395 {0.1337}
20	3.200 {0.1260}	41	3.410 {0.1343}
21	3.215 {0.1266}	42	3.425 {0.1348}
23	3.230 {0.1272}	44	3.440 {0.1354}
24	3.245 {0.1278}	45	3.455 {0.1360}
26	3.260 {0.1283}	47	3.470 {0.1366}
27	3.275 {0.1289}	–	–

(2) Press fit the bearing inner race (front) to the drive pinion gear using the **SSTs** and a press.

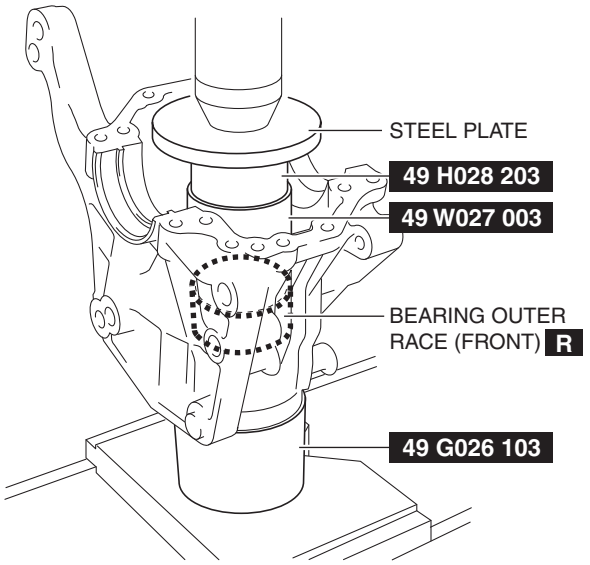
Caution
- **Do not apply pressure to the drive pinion teeth with the press. Otherwise, the drive pinion could be damaged.**

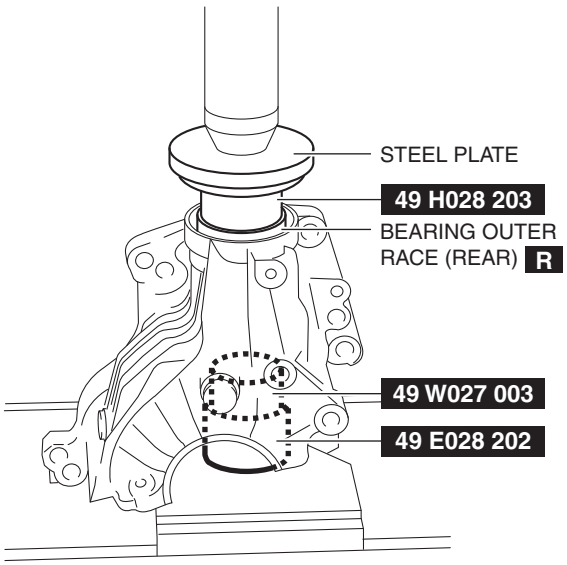
(3) Assemble a new collapsible spacer to the drive pinion gear component.



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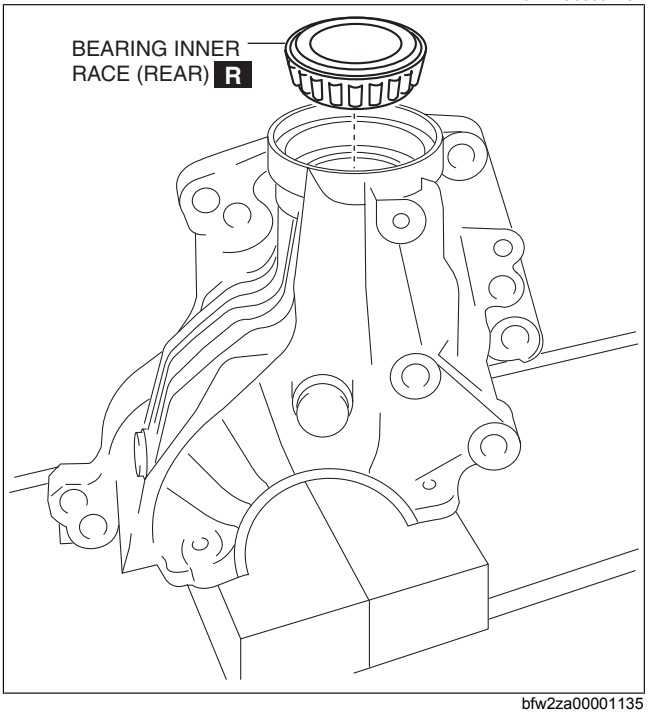
(4) Press fit the bearing outer race (front/rear) to the front carrier using the **SSTs** and a press, and a steel plate.





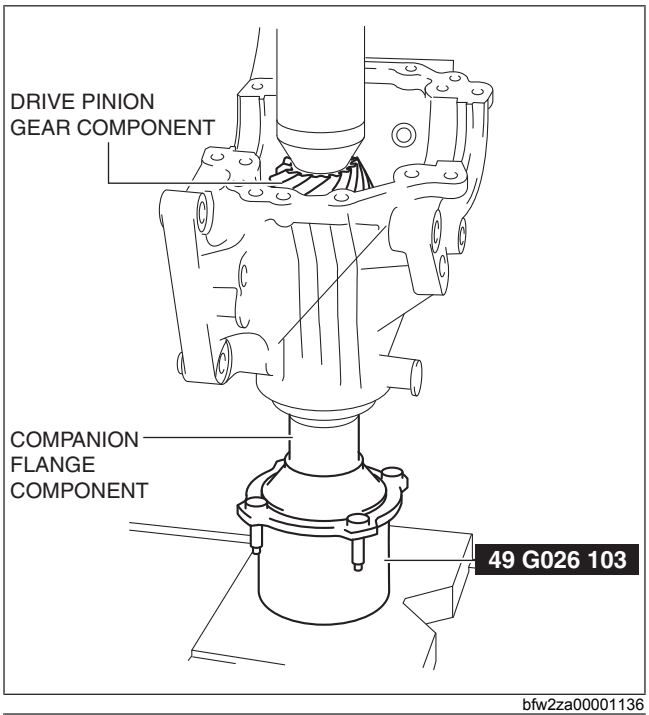
bfw2za00001134

(5) Assemble the bearing inner race (rear) to the front carrier.
(6) Apply grease to the area of the companion flange where it contacts the bearing inner race (rear).



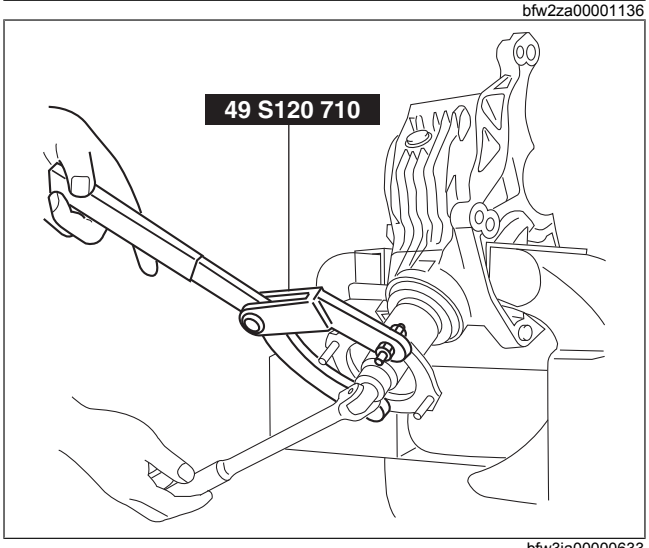
bfw2za00001135

(7) Assemble the drive pinion gear component and companion flange component to the front carrier using the **SST** and a press.
(8) Apply transfer oil to the thread area of the new locknut.



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(9) Assemble the washer and locknut to the drive pinion gear component using the **SST** and a vise.
(10) Rotate the companion flange by hand to seat the bearing.



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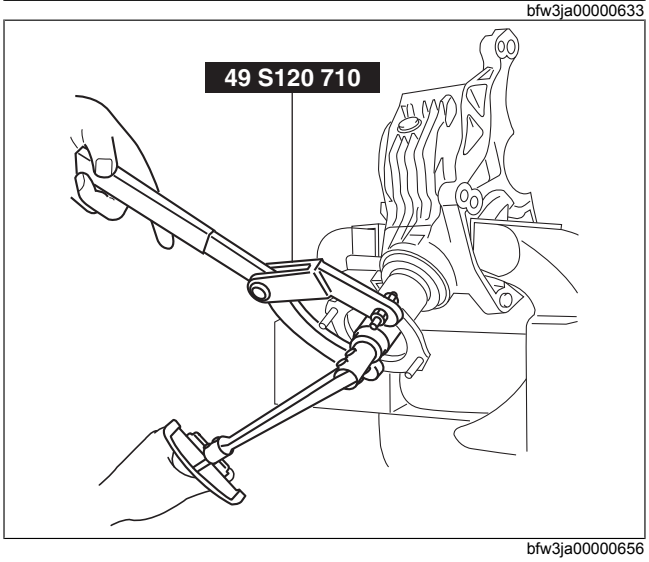
(11) Using the **SST** and a vise, gradually tighten the locknut from the minimum specified tightening torque to the specified preload. Record the tightening torque at this time.

Tightening torque
80—284 N·m {8.2—28 kgf·m, 60—209 ft·lbf}

Drive pinion preload
0.9—1.3 N·m {9.2—13 kgf·cm, 8.0—11 in·lbf}

- If the specified preload is not obtained within the specified tightening torque, replace the collapsible spacer with a new one and perform the preload adjustment again.

(12) Remove the following parts from the front carrier. (See TRANSFER DISASSEMBLY [EW6AX-EL].)
- Locknut
- Washer
- Companion flange component
- Drive pinion gear component
- Collapsible spacer

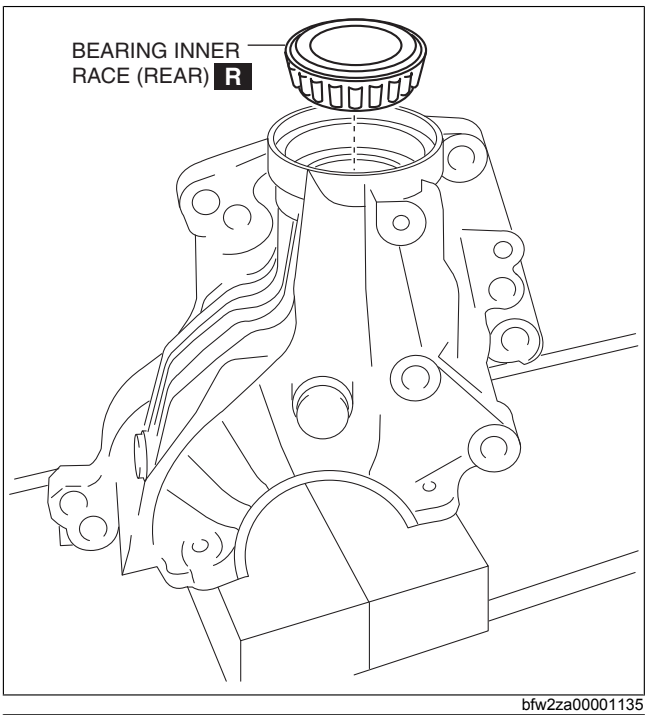


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- Bearing inner race (rear)

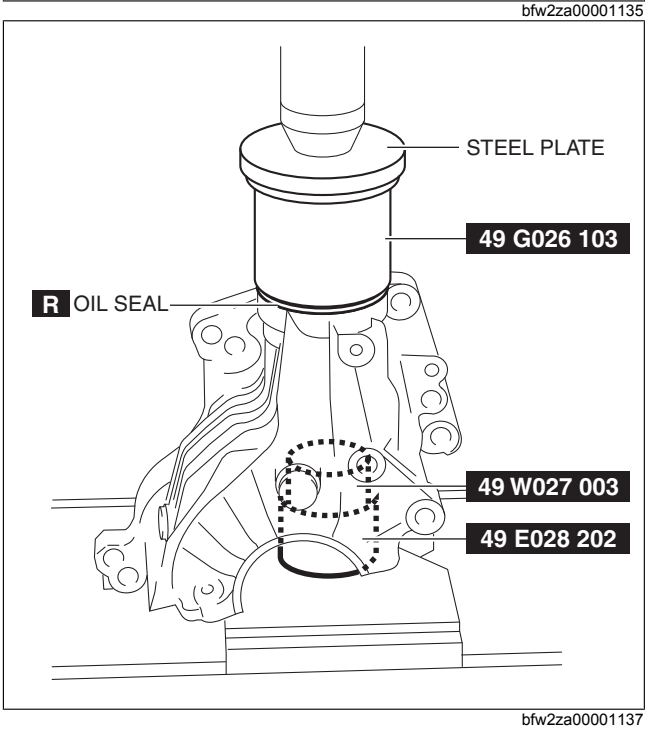
Note
- Reuse the collapsible spacer used in drive pinion preload measurement.

(13) Assemble the bearing inner race (rear) to the front carrier.



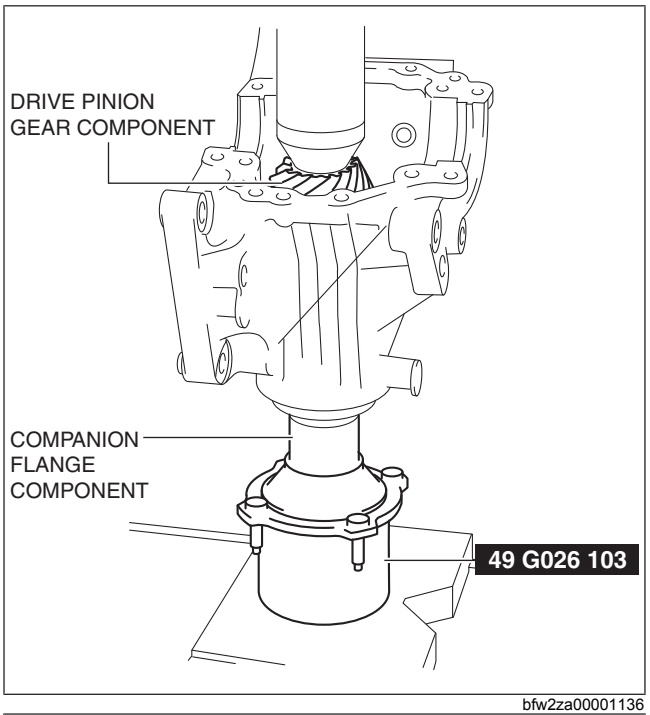
bfw2za00001135

(14) Assemble the new oil seal using the **SSTs** and a press, and a steel plate.
(15) Assemble the collapsible spacer used in drive pinion preload measurement to the drive pinion gear component.
(16) Apply grease to the area of the companion flange where it contacts the bearing inner race (rear).



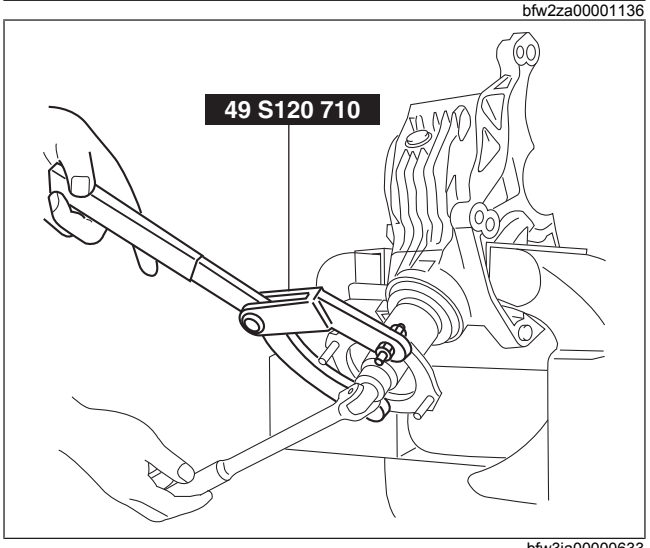
bfw2za00001137

(17) Assemble the drive pinion gear component and companion flange to the front carrier using the **SST** and a press.
(18) Apply transfer oil to the thread area of the new locknut.



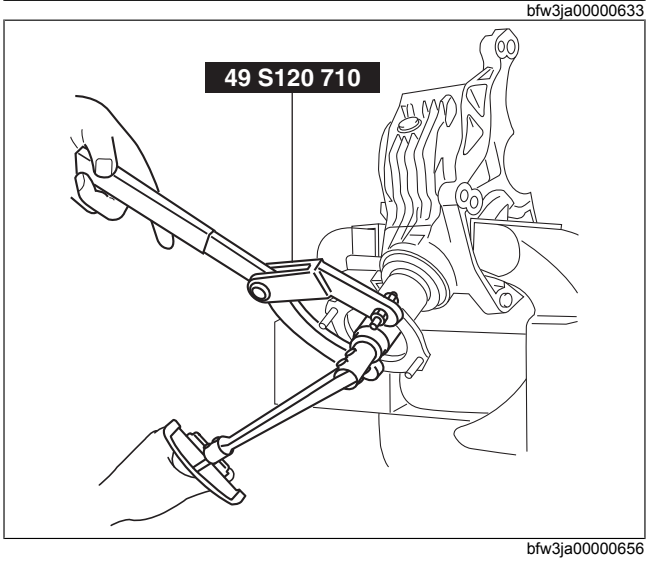
bfw2za00001136

(19) Assemble the washer and locknut to the drive pinion gear component using the **SST** and a vise.
(20) Rotate the companion flange by hand to seat the bearing.



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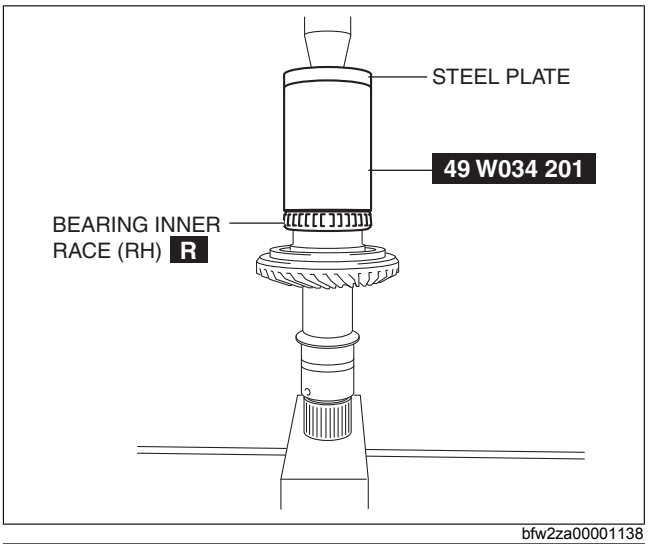
(21) Tighten the locknut to the torque recorded in drive pinion preload adjustment using the **SST** and a vise.



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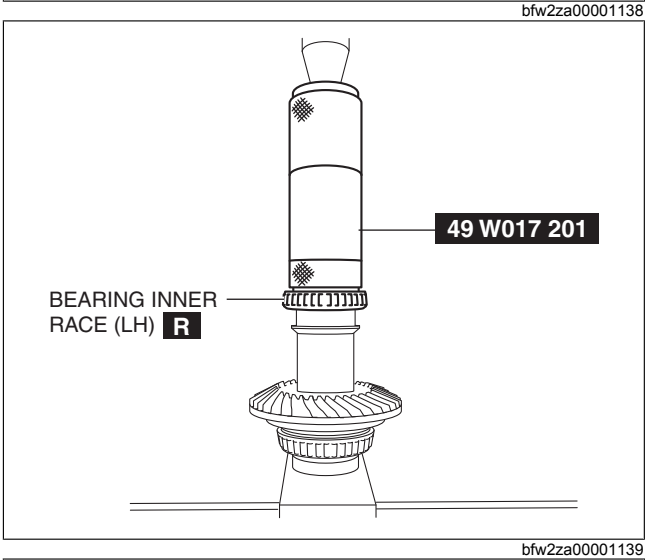
3. Assemble the ring gear shaft component to the front carrier using the following procedure.

(1) Press fit the bearing inner race (RH) to the ring gear shaft using the **SST** and a press, and a steel plate.



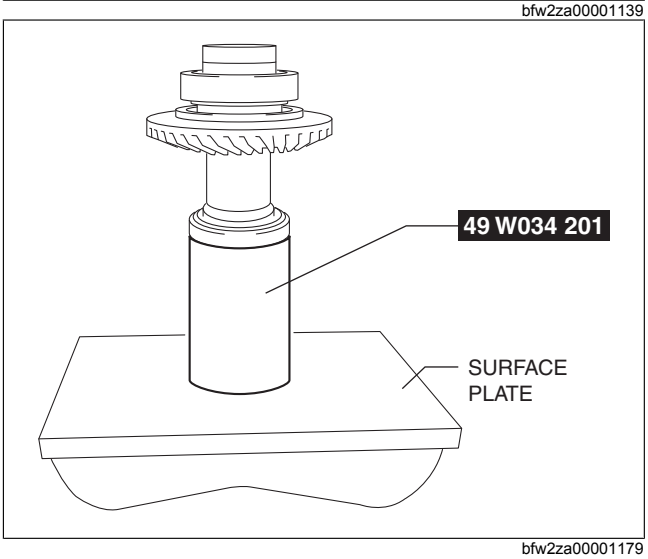
bfw2za00001138

(2) Press fit the bearing inner race (LH) to the ring gear shaft using the **SST** and a press.
(3) Temporarily assemble the bearing outer race (LH) and bearing outer race (RH) to the ring gear shaft.



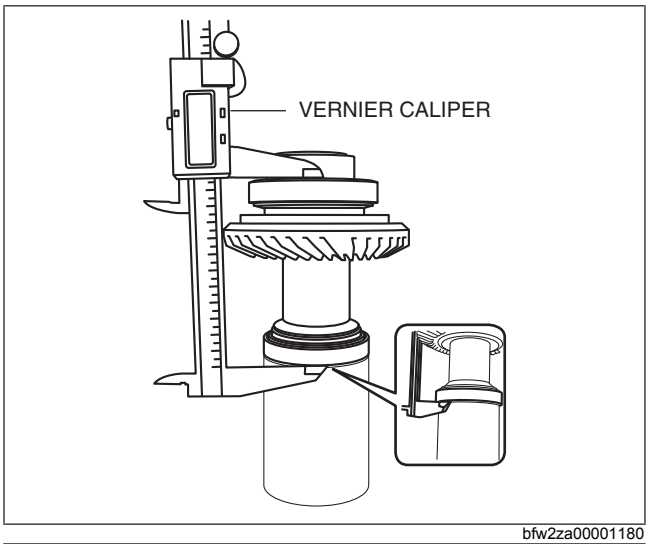
bfw2za00001139

(4) Place the **SST** on a surface plate as shown in the figure and set the ring gear shaft component on top of it with the bearing outer race (LH) side pointed downward.
(5) Rotate the bearing by hand several times to seat it.



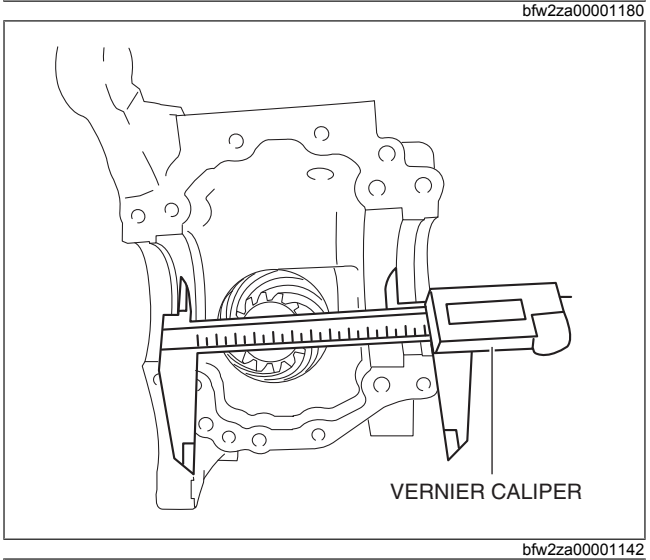
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(6) Using a vernier caliper, measure the distance between the bearing outer race (LH) and bearing outer race (RH).
   - Measure any 3 locations and designate the average value as distance A.



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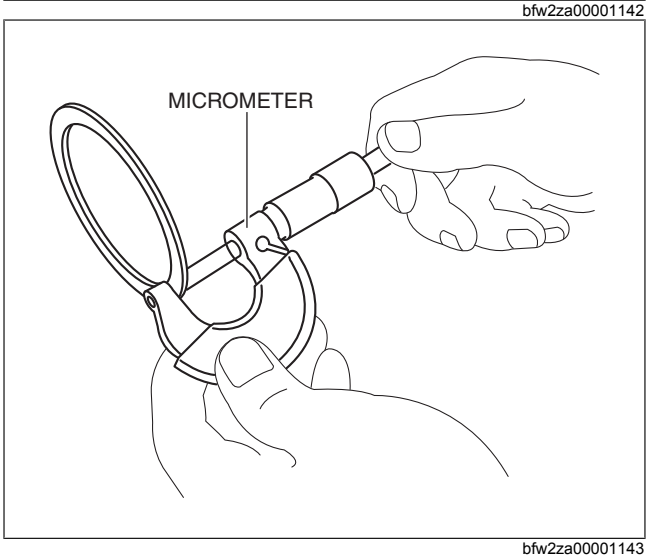
(7) Using a vernier caliper, measure the distance shown in the figure for the front carrier.
   - Measure any 3 locations and designate the average value as distance B.



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(8) Measure the thickness of adjustment shim (LH) and adjustment shim (RH) using a micrometer.
(9) Verify that the total value (thickness) of adjustment shim (LH) and adjustment shim (RH) is within the range of the minimum and maximum values.

Minimum value = Distance B - Distance A + 0.07 mm {0.003 in}
Maximum value = Distance B - Distance A + 0.16 mm {0.0063 in}



bfw2za00001143

   - If the total value (thickness) of adjustment shim (LH) and adjustment shim (RH) is within the minimum and maximum values, use them as is.
   - If the total thickness of adjustment shim (LH) and adjustment shim (RH) is not within the range of the maximum and minimum values, select the appropriate adjustment shims from the following table.

Adjustment shim (LH) table

Part number	Thickness (mm {in})	Part number	Thickness (mm {in})
CN01 27 3D1	2.490 {0.09803}	CN01 27 3F1	3.030 {0.1193}
CN01 27 3D2	2.520 {0.09921}	CN01 27 3F2	3.060 {0.1205}
CN01 27 3D3	2.550 {0.1004}	CN01 27 3F3	3.090 {0.1217}
CN01 27 3D4	2.580 {0.1016}	CN01 27 3F4	3.120 {0.1228}
CN01 27 3D5	2.610 {0.1028}	CN01 27 3F5	3.150 {0.1240}
CN01 27 3D6	2.640 {0.1039}	CN01 27 3F6	3.180 {0.1252}

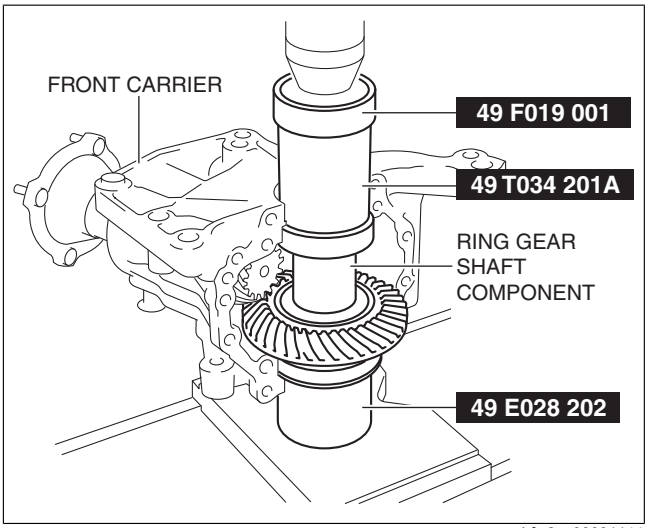
Part number	Thickness (mm {in})	Part number	Thickness (mm {in})
CN01 27 3D7	2.670 {0.1051}	CN01 27 3F7	3.210 {0.1264}
CN01 27 3D8	2.700 {0.1063}	CN01 27 3F8	3.240 {0.1276}
CN01 27 3D9	2.730 {0.1075}	CN01 27 3F9	3.270 {0.1287}
CN01 27 3E1	2.760 {0.1087}	CN01 27 3G1	3.300 {0.1299}
CN01 27 3E2	2.790 {0.1098}	CN01 27 3G2	3.330 {0.1311}
CN01 27 3E3	2.820 {0.1110}	CN01 27 3G3	3.360 {0.1323}
CN01 27 3E4	2.850 {0.1122}	CN01 27 3G4	3.390 {0.1335}
CN01 27 3E5	2.880 {0.1134}	CN01 27 3G5	3.420 {0.1346}
CN01 27 3E6	2.910 {0.1146}	CN01 27 3G6	3.450 {0.1358}
CN01 27 3E7	2.940 {0.1157}	CN01 27 3G7	3.480 {0.1370}
CN01 27 3E8	2.970 {0.1169}	CN01 27 3G8	3.510 {0.1382}
CN01 27 3E9	3.000 {0.1181}	–	–

Adjustment shim (RH) table

Part number	Thickness (mm {in})	Part number	Thickness (mm {in})
CN01 27 355	2.490 {0.09803}	CN01 27 375	3.030 {0.1193}
CN01 27 356	2.520 {0.09921}	CN01 27 376	3.060 {0.1205}
CN01 27 357	2.550 {0.1004}	CN01 27 377	3.090 {0.1217}
CN01 27 358	2.580 {0.1016}	CN01 27 378	3.120 {0.1228}
CN01 27 359	2.610 {0.1028}	CN01 27 379	3.150 {0.1240}
CN01 27 361	2.640 {0.1039}	CN01 27 3B1	3.180 {0.1252}
CN01 27 362	2.670 {0.1051}	CN01 27 3B2	3.210 {0.1264}
CN01 27 363	2.700 {0.1063}	CN01 27 3B3	3.240 {0.1276}
CN01 27 364	2.730 {0.1075}	CN01 27 3B4	3.270 {0.1287}
CN01 27 365	2.760 {0.1087}	CN01 27 3B5	3.300 {0.1299}
CN01 27 366	2.790 {0.1098}	CN01 27 3B6	3.330 {0.1311}
CN01 27 367	2.820 {0.1110}	CN01 27 3B7	3.360 {0.1323}
CN01 27 368	2.850 {0.1122}	CN01 27 3B8	3.390 {0.1335}
CN01 27 369	2.880 {0.1134}	CN01 27 3B9	3.420 {0.1346}
CN01 27 371	2.910 {0.1146}	CN01 27 3C1	3.450 {0.1358}
CN01 27 372	2.940 {0.1157}	CN01 27 3C2	3.480 {0.1370}
CN01 27 373	2.970 {0.1169}	CN01 27 3C3	3.510 {0.1382}
CN01 27 374	3.000 {0.1181}	–	–

(10) Assemble the following parts to the front carrier using the **SSTs** and a press.
- Ring gear shaft
- Bearing outer race (LH)
- Bearing outer race (RH)
- Adjustment shim (LH)
- Adjustment shim (RH)

Caution
- **Do not apply pressure of 2 t or more with the press. Otherwise, it could damage the part.**



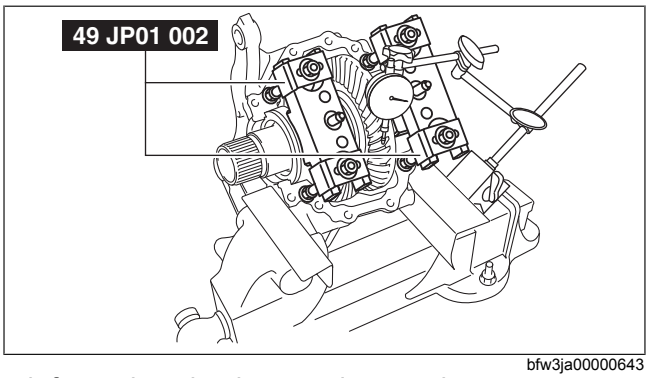
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4. Perform the ring gear and drive pinion backlash adjustment.

Note
- Install the **SSTs** so that they do not contact the ring gear.

(1) Install the **SSTs** and dial gauge to the differential carrier using a vise as shown in the figure.

SST tightening torque
5.0 N·m {51 kgf·cm, 44 in·lbf}

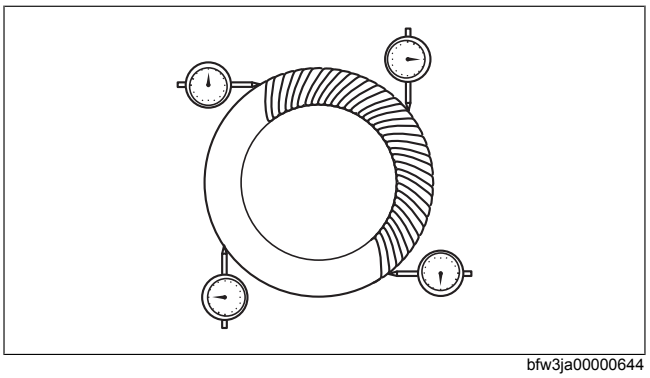


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(2) Secure the drive pinion gear and measure the backlash from when the ring gear is moved.

Drive pinion gear and ring gear backlash
Standard: 0.09—0.14 mm {0.004—0.005 in}
Minimum: 0.05 mm {0.002 in}
Variance: 0.07 mm {0.003 in}

Note
- Measure the backlash at 4 locations around the ring gear. Verify that 1 of the 4 locations is within the standard, and the minimum value for the 4 locations is **0.05 mm or more**, and the variance is **0.07 mm or less**.
  - If the measured value of the backlash is not within the standard, adjust it by moving the ring gear shaft in the axial direction.


bfw3ja00000644

Note
- Replace the adjustment shims to move the ring gear shaft in the axial direction. At this time, if the adjustment shim on the right side is replaced with one **0.03 mm** thicker, replace the adjustment shim on the left side with one **0.03 mm** thinner.

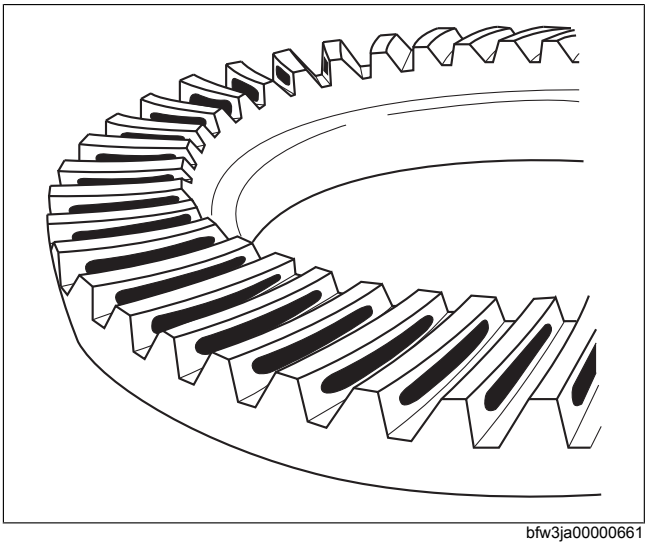
Adjustment shim (LH) table

Part number	Thickness (mm {in})	Part number	Thickness (mm {in})
CN01 27 3D1	2.490 {0.09803}	CN01 27 3F1	3.030 {0.1193}
CN01 27 3D2	2.520 {0.09921}	CN01 27 3F2	3.060 {0.1205}
CN01 27 3D3	2.550 {0.1004}	CN01 27 3F3	3.090 {0.1217}
CN01 27 3D4	2.580 {0.1016}	CN01 27 3F4	3.120 {0.1228}
CN01 27 3D5	2.610 {0.1028}	CN01 27 3F5	3.150 {0.1240}
CN01 27 3D6	2.640 {0.1039}	CN01 27 3F6	3.180 {0.1252}
CN01 27 3D7	2.670 {0.1051}	CN01 27 3F7	3.210 {0.1264}
CN01 27 3D8	2.700 {0.1063}	CN01 27 3F8	3.240 {0.1276}
CN01 27 3D9	2.730 {0.1075}	CN01 27 3F9	3.270 {0.1287}
CN01 27 3E1	2.760 {0.1087}	CN01 27 3G1	3.300 {0.1299}
CN01 27 3E2	2.790 {0.1098}	CN01 27 3G2	3.330 {0.1311}
CN01 27 3E3	2.820 {0.1110}	CN01 27 3G3	3.360 {0.1323}
CN01 27 3E4	2.850 {0.1122}	CN01 27 3G4	3.390 {0.1335}
CN01 27 3E5	2.880 {0.1134}	CN01 27 3G5	3.420 {0.1346}
CN01 27 3E6	2.910 {0.1146}	CN01 27 3G6	3.450 {0.1358}
CN01 27 3E7	2.940 {0.1157}	CN01 27 3G7	3.480 {0.1370}
CN01 27 3E8	2.970 {0.1169}	CN01 27 3G8	3.510 {0.1382}
CN01 27 3E9	3.000 {0.1181}	–	–

Adjustment shim (RH) table

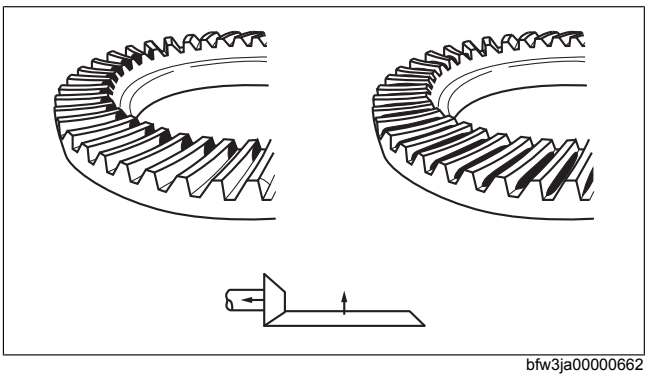
Part number	Thickness (mm {in})	Part number	Thickness (mm {in})
CN01 27 355	2.490 {0.09803}	CN01 27 375	3.030 {0.1193}
CN01 27 356	2.520 {0.09921}	CN01 27 376	3.060 {0.1205}
CN01 27 357	2.550 {0.1004}	CN01 27 377	3.090 {0.1217}
CN01 27 358	2.580 {0.1016}	CN01 27 378	3.120 {0.1228}
CN01 27 359	2.610 {0.1028}	CN01 27 379	3.150 {0.1240}
CN01 27 361	2.640 {0.1039}	CN01 27 3B1	3.180 {0.1252}
CN01 27 362	2.670 {0.1051}	CN01 27 3B2	3.210 {0.1264}
CN01 27 363	2.700 {0.1063}	CN01 27 3B3	3.240 {0.1276}
CN01 27 364	2.730 {0.1075}	CN01 27 3B4	3.270 {0.1287}
CN01 27 365	2.760 {0.1087}	CN01 27 3B5	3.300 {0.1299}
CN01 27 366	2.790 {0.1098}	CN01 27 3B6	3.330 {0.1311}
CN01 27 367	2.820 {0.1110}	CN01 27 3B7	3.360 {0.1323}
CN01 27 368	2.850 {0.1122}	CN01 27 3B8	3.390 {0.1335}
CN01 27 369	2.880 {0.1134}	CN01 27 3B9	3.420 {0.1346}
CN01 27 371	2.910 {0.1146}	CN01 27 3C1	3.450 {0.1358}
CN01 27 372	2.940 {0.1157}	CN01 27 3C2	3.480 {0.1370}
CN01 27 373	2.970 {0.1169}	CN01 27 3C3	3.510 {0.1382}
CN01 27 374	3.000 {0.1181}	–	–

5. Perform the drive pinion gear and ring gear shaft tooth contact inspection using the following procedure.
   (1) Apply tooth marking compound (red lead coating) evenly to both tooth surfaces of the ring gear.
   (2) Rotate the ring gear back and forth several times.
   (3) Inspect the tooth contact pattern in 4 locations around the ring gear and verify that the tooth contact points exhibit the pattern shown in the figure.



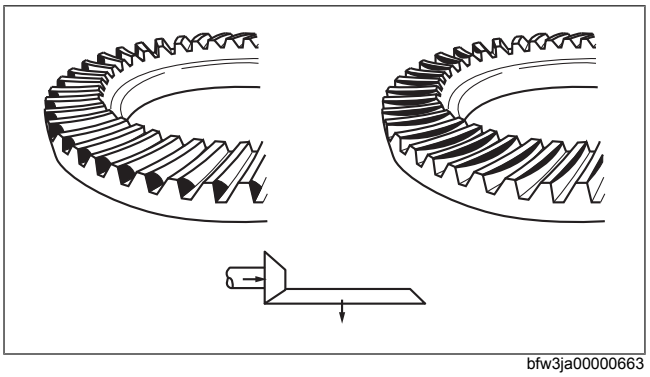
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- If the tooth contact points are normal, wipe off the marking compound.
- If the tooth contact is not correct, adjust the drive pinion gear spacer. Then, adjust the drive pinion preload and the backlash.
  — If the tow and flank contact points appear as shown in the figure, replace the drive pinion gear spacer with a thinner one to move the drive pinion gear farther away from the ring gear.



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— If the heel and face contact points appear as shown in the figure, replace the drive pinion gear spacer with a thicker one to move the drive pinion gear closer to the ring gear.



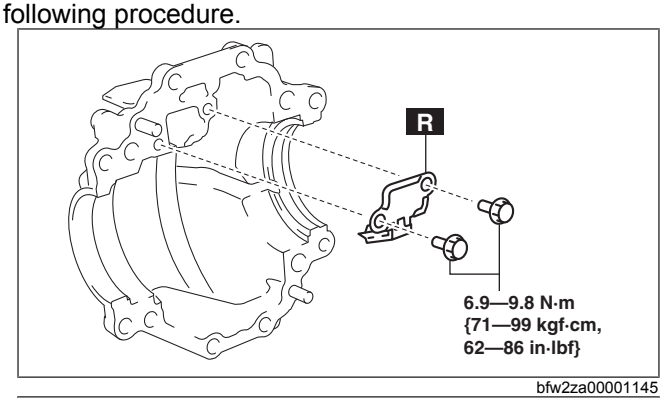
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Spacer table

Identification number	Thickness (mm {in})	Identification number	Thickness (mm {in})
08	3.080 {0.1213}	29	3.290 {0.1295}
09	3.095 {0.1219}	30	3.305 {0.1301}
11	3.110 {0.1224}	32	3.320 {0.1307}
12	3.125 {0.1230}	33	3.335 {0.1313}
14	3.140 {0.1236}	35	3.350 {0.1319}
15	3.155 {0.1242}	36	3.365 {0.1325}
17	3.170 {0.1248}	38	3.380 {0.1331}
18	3.185 {0.1254}	39	3.395 {0.1337}
20	3.200 {0.1260}	41	3.410 {0.1343}
21	3.215 {0.1266}	42	3.425 {0.1348}
23	3.230 {0.1272}	44	3.440 {0.1354}
24	3.245 {0.1278}	45	3.455 {0.1360}
26	3.260 {0.1283}	47	3.470 {0.1366}
27	3.275 {0.1289}	–	–

6. Install the drive gear case to the front carrier using the following procedure.
   (1) Assemble a new baffle plate to the drive gear case.

   Tightening torque
   6.9—9.8 N·m {71—99 kgf·cm, 62—86 in·lbf}

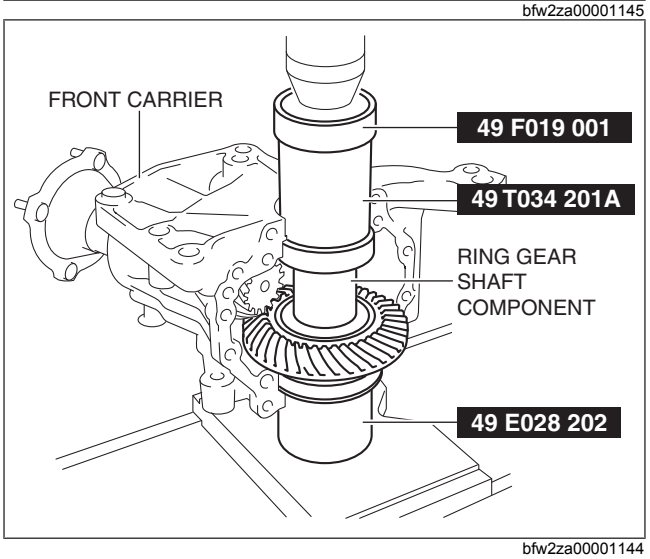


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   (2) Verify that there is no gap between the baffle plate seal area lip and the drive gear case.

   (3) Apply pressure to the ring gear shaft component using the **SSTs** and a press.

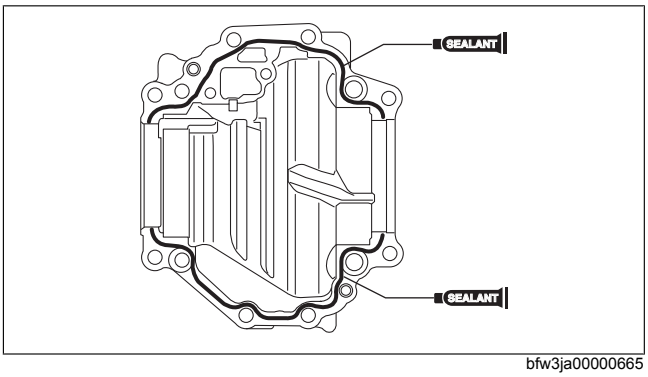
   Caution
   - **Do not apply pressure of 2 t or more with the press. Otherwise, it could damage the part.**



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(4) Apply a thin layer of sealant (silicone sealant TB1217C or equivalent) to the locations indicated in the figure for the drive gear case.

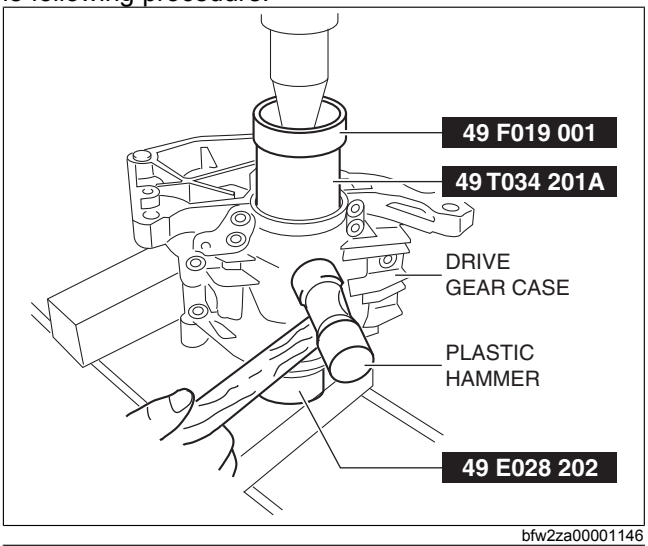
Caution
- **To prevent the silicone sealant from hardening, adhere the drive gear case to the front carrier within 3 min after applying silicone sealant.**



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(5) Assemble the drive gear case to the front carrier.

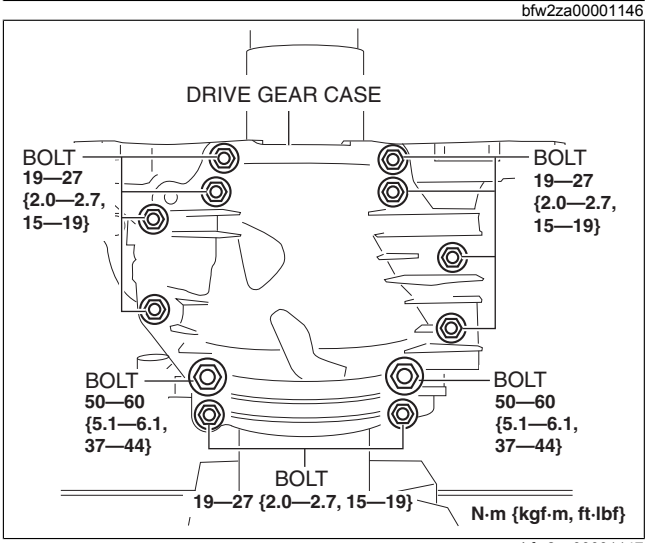
Note
- If the drive gear case is difficult to assemble, use the following procedure.
  1. Tap the drive gear case with a plastic hammer.
  2. Temporarily tighten the drive gear case installation bolts.
  3. Release the pressure on the press.
  4. Adjust the position of the **SSTs** and apply pressure to the press again.



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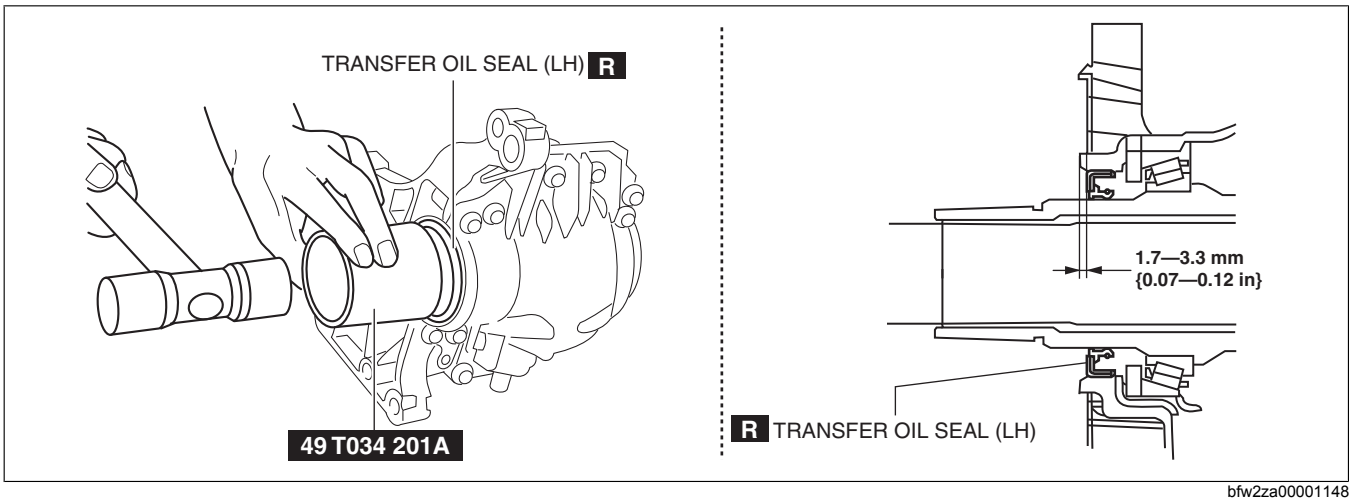
(6) Tighten the drive gear case installation bolts to the specified torque.
(7) Assemble a new breather.

Tightening torque
8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}



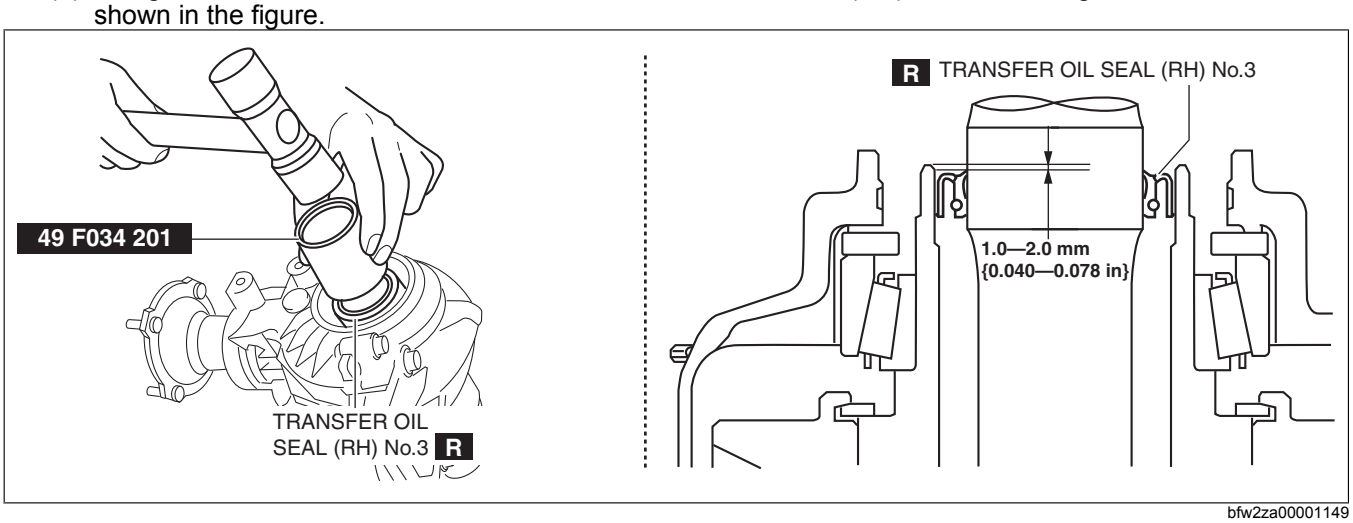
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7. Assemble a transfer oil seal using the following procedure.
   (1) Using the **SST** and a hammer, assemble a new transfer oil seal (LH) according to the dimensions shown in the figure.



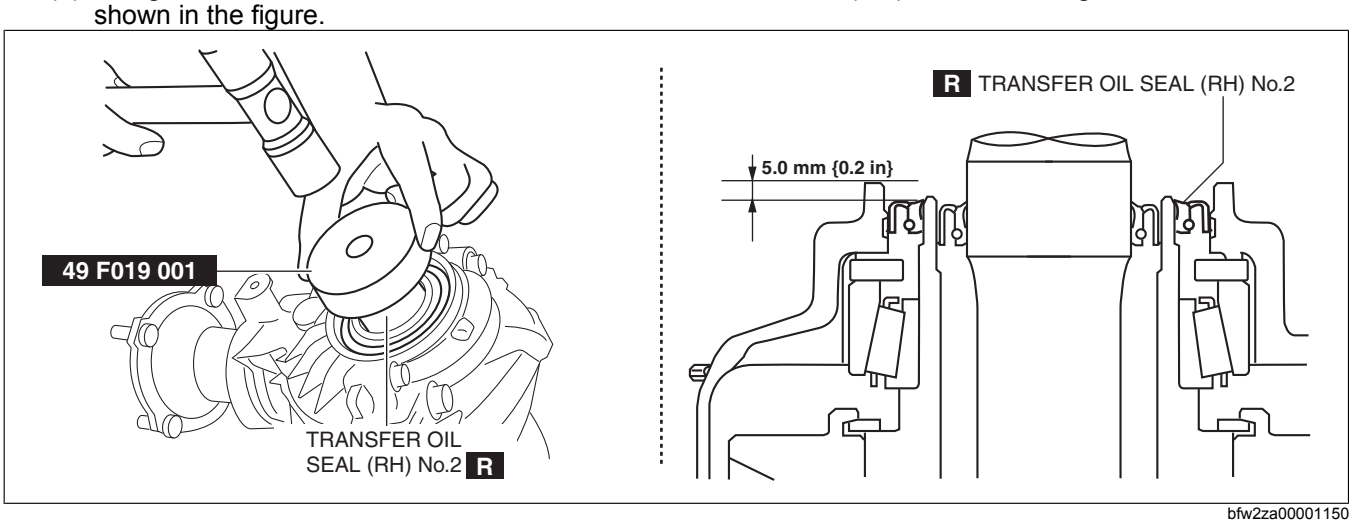
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(2) Using the **SST** and a hammer, assemble a new transfer oil seal (RH) No.3 according to the dimensions shown in the figure.



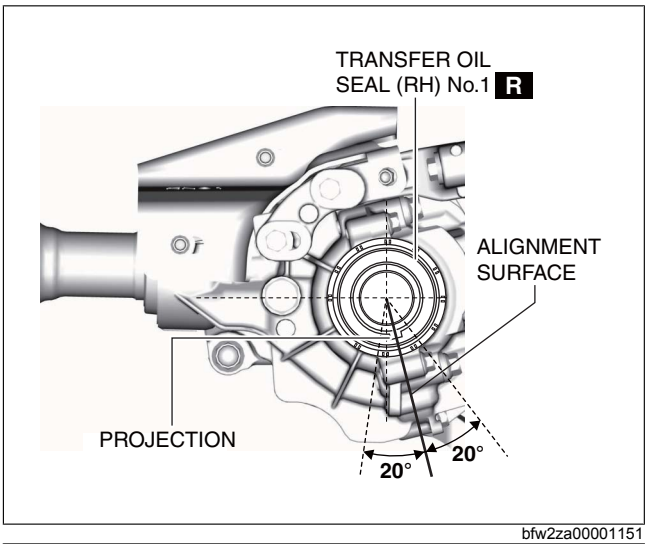
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(3) Using the **SST** and a hammer, assemble a new transfer oil seal (RH) No.2 according to the dimensions shown in the figure.



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(4) Install the transfer oil seal (RH) No.1 so that the projection of the transfer oil seal (RH) No.1 is within the range shown in the figure. (With projection)



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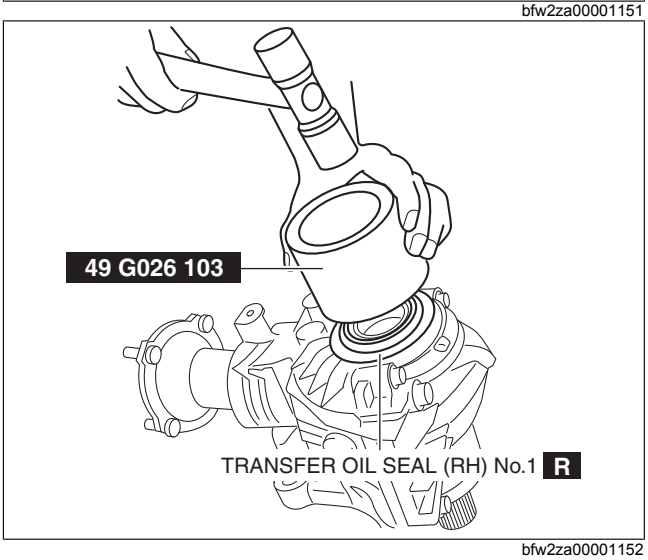
(5) Using the **SST** and a hammer, assemble a new transfer oil seal (RH) No.1.

8. Assemble a new washer and drain plug.

Tightening torque
40—58 N·m {4.1—5.9 kgf·m, 30—42 ft·lbf}

9. Assemble a new washer and oil level plug.

Tightening torque
40—58 N·m {4.1—5.9 kgf·m, 30—42 ft·lbf}



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